MongoDB Practical Assessment 2 — Food Delivery

Tools: MongoDB Compass + `mongosh`

Database: `food_delivery_assessment`

Data Setup

```
use food_delivery_assessment
```

Restaurants

```
db.restaurants.insertMany([
    {
        restaurant_id: 101,
        restaurant_name: "Spice Kitchen",
        city: "Hyderabad",
        cuisine: ["Indian", "Biryani"],
        rating: 4.5,
        active: true,
        contact: {
            phone: "9876543210",
            email: "spice@email.com"
        }
    },
    {
        restaurant_id: 102,
        restaurant_name: "Urban Pizza",
        city: "Mumbai",
        cuisine: ["Italian", "Fast Food"],
        rating: 4.2,
        active: true,
        contact: {
            phone: "9988776655",
            email: "pizza@email.com"
        }
```

```
    },
    {
        restaurant_id: 103,
        restaurant_name: "Green Bowl",
        city: "Bangalore",
        cuisine: ["Healthy", "Vegetarian"],
        rating: 4.7,
        active: true,
        contact: {
            phone: "9123456780",
            email: "green@email.com"
        }
    },
    {
        restaurant_id: 104,
        restaurant_name: "Burger Point",
        city: "Hyderabad",
        cuisine: ["Fast Food", "Burgers"],
        rating: 3.9,
        active: false,
        contact: {
            phone: "9000011111",
            email: "burger@email.com"
        }
    },
    {
        restaurant_id: 105,
        restaurant_name: "Asian Wok",
        city: "Delhi",
        cuisine: ["Chinese", "Thai"],
        rating: 4.4,
        active: true,
        contact: {
            phone: "9222233333",
            email: "wok@email.com"
        }
    }
])
```

Orders

```
db.orders.insertMany([
    {
```

```
    order_id: 5001,
    restaurant_id: 101,
    customer_name: "Adil",
    order_date: ISODate("2026-09-01"),
    status: "Delivered",
    items: [
        { name: "Chicken Biryani", quantity: 2, price: 320 },
        { name: "Coke", quantity: 2, price: 60 }
    ],
    delivery: {
        city: "Hyderabad",
        fee: 40
    }
},
{
    order_id: 5002,
    restaurant_id: 102,
    customer_name: "Megha",
    order_date: ISODate("2026-09-02"),
    status: "Processing",
    items: [
        { name: "Farmhouse Pizza", quantity: 1, price: 550 }
    ],
    delivery: {
        city: "Mumbai",
        fee: 60
    }
},
{
    order_id: 5003,
    restaurant_id: 103,
    customer_name: "Kabir",
    order_date: ISODate("2026-09-03"),
    status: "Delivered",
    items: [
        { name: "Protein Bowl", quantity: 2, price: 400 },
        { name: "Juice", quantity: 1, price: 150 }
    ],
    delivery: {
        city: "Bangalore",
        fee: 50
    }
},
{
```

```
        order_id: 5004,
        restaurant_id: 101,
        customer_name: "Nisha",
        order_date: ISODate("2026-09-04"),
        status: "Cancelled",
        items: [
            { name: "Veg Biryani", quantity: 1, price: 250 }
        ],
        delivery: {
            city: "Hyderabad",
            fee: 40
        }
    },
    {
        order_id: 5005,
        restaurant_id: 105,
        customer_name: "Rohan",
        order_date: ISODate("2026-09-05"),
        status: "Delivered",
        items: [
            { name: "Thai Curry", quantity: 1, price: 480 },
            { name: "Noodles", quantity: 2, price: 300 }
        ],
        delivery: {
            city: "Delhi",
            fee: 70
        }
    }
])
```

Exercises

CRUD and Basic Queries

1. Display all restaurants.
2. Find only restaurants from Hyderabad.
3. Find active restaurants having rating greater than `4.3`.
4. Insert one new restaurant of your choice.
5. Change the rating of restaurant `104`.
6. Change restaurant `104` from inactive to active.
7. Delete the restaurant you inserted in Question 4.

Arrays and Embedded Documents

8. Find restaurants serving `Fast Food`.
9. Find restaurants serving both `Italian` and `Fast Food`.
10. Add `Desserts` to the cuisine list of restaurant `102`.
11. Change the email address of restaurant `103`.
12. Display restaurants whose contact phone starts with `9`.

Orders

13. Find all Delivered orders.
14. Find orders placed after `2026-09-02`.
15. Find orders delivered to Hyderabad.
16. Find orders containing an item named `Noodles`.
17. Find orders containing an item where quantity is greater than `1`.
18. Find an order where the **same item** has:

```
name = Chicken Biryani
quantity = 2
```

Use an appropriate array operator.

19. Add another item to order `5002`.
20. Change order `5002` from `Processing` to `Delivered`.

Projection and Sorting

21. Display only:

```
restaurant_name
city
rating
```

without `_id`.

22. Sort restaurants by rating from highest to lowest.
23. Display the top 3 highest-rated restaurants.

Data Types

24. Find documents where `rating` is stored as a numeric value.
25. Identify the datatype of `order_date` .
26. Identify the datatype of `items` .
27. Insert one test restaurant where `rating` is accidentally stored as a string, identify it using `$type` , and then correct the value.

Final Challenge

28. Create a new collection called:

```
delivery_partners
```

Create at least 4 documents containing:

```
partner_id
partner_name
city
rating
active
vehicle
skills/areas as an array
contact as an embedded document
```

Then:

29. Find active delivery partners from Hyderabad.
30. Find partners with rating greater than `4` .
31. Add a new service area to one partner.
32. Update one partner's rating.
33. Sort all partners from highest rating to lowest.